East West University

Department of Computer Science and Engineering

Semester : spring 2021

Course Title: Electrical Circuits

Course Code: CSE 209

Section: 03

Experiment No: 02

Experiment Title: Series – Parallel DC Circuit and Verification of Kirchhoff'S Laws

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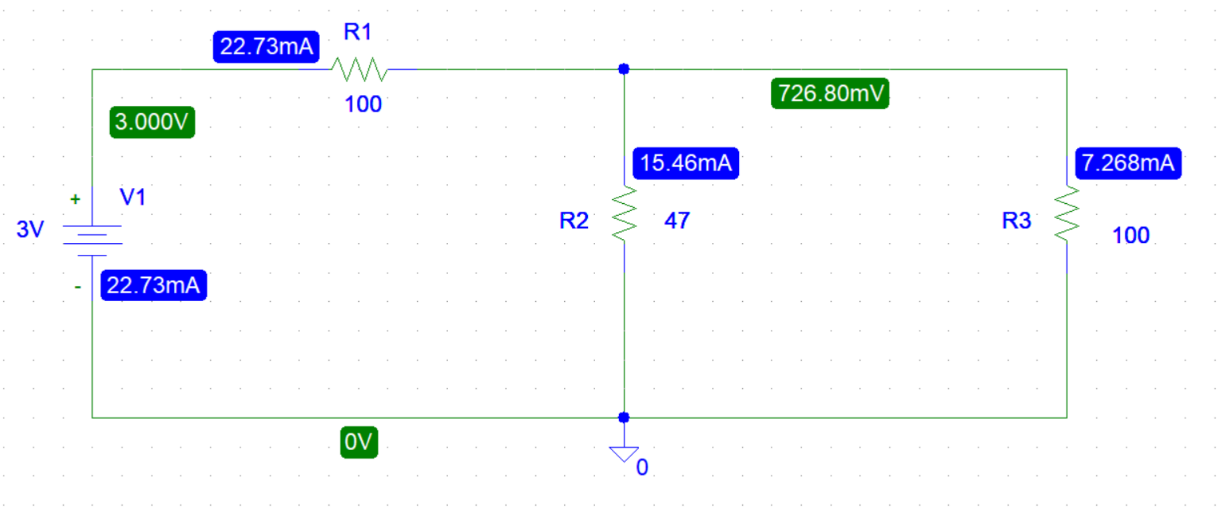
Date of Performance: 22.03.2021

Date of Submission: 28.03.2021

Objectives:

1. To learn analysis of dc series-parallel circuit.
2. To verify Kirchhoff's voltage law (KVL).
3. To verify Kirchhoff's current law (KCL).

Circuit Diagram:



Experimental Datasheet:

Measured Value of E(V)	Measured Value of V_1 (V)	Measured Value of V_2 (V)	Measured Value of V_3 (V)	Measured Value of I_1 (mA)	Measured Value of I_2 (mA)	Measured Value of I_3 (mA)	Measured Value of Resistance(Ω)
3	3	0.7268	0.7268	22.73	15.46	7.268	R_1 =100 Ω R_2 =47Ω R_3 =100 Ω

Answers to the post lab report questions:

1. Here , R_2 and R_3 are parallel,

So, Equivalent resistance, Rp= $\frac{47\times100}{47+100} = 31.973\ \Omega$

Applying voltage divider rule,

$V_1 = \frac{E}{R1+Rp} \times R_1$

$= \frac{3}{100+31.973} \times 100$

= 2.273V

$V_2 = \frac{E}{R1+Rp} \times Rp$

$= \frac{3}{100+31.973} \times 31.973$

= 0.727V

Here , V_2 and V_3 are parallel.

So , $V_2 = V_3$ = 0.727V . Because the voltage is equal in Parallel Circuit .

From Ohm's Law,

$I_1 = \frac{E}{R1+Rp}$

$= \frac{3}{100+31.973}$

= 0.02273A

= 22.73mA

Applying current divider rule,

$I_2 = \frac{I1}{R2+R3} \times R_3$

$= \frac{22.73}{47+100} \times 100$

= 15.463mA

$I_3 = \frac{I1}{R2+R3} \times R_2$

$= \frac{22.73}{47+100} \times 47$

= 7.267mA

Comparing the theoretically calculated values with the simulated values from PSpice.

	Calculated value	Measured value
Current	I_1 =22.73mA	I_1 =22.73mA
	I_2= 15.463mA	I_2 =15.46mA
	I_3= 7.267mA	I_3= 7.268mA
Voltage	V_1 = 2.273V	V_1 = 3V
	V_2 = 0.727V = 727mV	V_2 = 726.80mV
	V_3 = 0.727V = 727mV	V_3 = 726.80mV

The experimental values of the voltages and the currents are slightly different from the theoretical values.

2. (i) From the calculated values we get,

$$V_2 = V_3 = 0.727\text{V}$$

Here, both V_2 and V_3 are parallel in the circuit so the voltage are same.

(ii) From the calculated values we get,

$V_1 = 2.273V$

$V_2 = 0.727V$

Applying KVL,

$$V_1+V_2 = (2.273+0.727)V$$
$$= 3V$$
$$= E$$

(iii) From the calculated values we get,

$I_1 = 22.73mA$

$I_2 = 15.463mA$

$I_3 = 7.267mA$

Applying KCL,

$$I_2 + I_3 = (15.463+7.267)mA$$
$$= 22.73mA$$
$$= I_1$$

Conclusion:

From the experiment we learnt about the application Ohm's Law, Kirchhoff's Voltage Law(KVL) and Kirchhoff's Current Law(KCL).